

A new model for a photovoltaic panel using Proteus software tool under arbitrary environmental conditions

Salam J. Yaqoob^{a,*}, Saad Motahhir^b, Ephraim Bonah Agyekum^c

^a Department of Research and Education, Authority of the Popular Crowd, Prime Minister, Baghdad, 10001, Iraq

^b Engineering, Systems and Applications Laboratory, ENSA, SMBA University, Fez, 30000, Morocco

^c Department of Nuclear and Renewable Energy, Ural Federal University named after the first President of Russia Boris, Yeltsin, 620002, 19 Mira Street, Ekaterinburg, Russia

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ABSTRACT

The photovoltaic (PV) panel generates power based on different parameters, including environmental conditions such as solar irradiance, temperature, and internal electrical parameters of the PV panel. Thus, a PV model should be studied in advance to forecast and evaluate the impact of these factors on the PV performance, and this model should be matched with the PV panel's real behavior. For this reason, this paper developed a new model for a PV panel using the Proteus software. A flexible PV model with possibility of varying the weather conditions has been proposed using mathematical equations of a single-diode equivalent circuit, (i.e., photo-generated current, shunt-resistor current, and diode current). Moreover, these equations were modeled using "Pick Devices" in Proteus software library. To prove the effectiveness of the proposed model, it was applied on two types of monocrystalline, and multi-crystalline silicon cells of PV panel technologies. Subsequently, the corresponding current-voltage (I–V) and power-voltage (P–V) characteristics are obtained and compared with the manufacturer's data, PVsyst software, and other authors' results under arbitrary values of irradiance and temperature. The simulation results identified that the proposed model is flexible, simple, and suitable for different PV panel technologies under diverse weather conditions.

1. Introduction

Nowadays, photovoltaic (PV) panel-based renewable energy harvesting is one of the most important energy sources that is used globally due to its high availability (Volker, 2005). A PV cell converts solar energy directly into electrical energy by a physical process called the photoelectric effect (Agyekum, 2021). Besides, the PV cell has current-voltage (I–V) characteristics similar to an exponential behavior of a PN junction, which means there is an interface between the two semiconductor material types, namely the P-type and the N-type inside a semiconductor (Adawi, 1964). Since the open-circuit voltage of the PV cell depends on the semiconductor gap and not on the size, there will always be a 0.6 V across an open-circuited cell (Walker, 2001). Therefore, many cells have to be connected in series to form a PV panel which can then be connected to form a larger unit called an array (Kawamura et al., 2003). Generally, the PV panel or cell circuit model must be studied to analyze its electrical behavior (Yahya-Khotbehsara and Shahhoseini, 2018). Hence, the electrical equivalent-circuit is used to

represent a PV panel or cell (Keevers and Green, 1994).

In literature, several researchers have studied the PV panel circuit models, such as single-diode and double-diode models (Lo Brano et al., 2010). Anani and Ibrahim (2020) used a numerical technique to compute the lumped-circuit parameters of a single diode model under STC conditions. The Matlab/Simulink was used to test the performance of the proposed method under different weather conditions. Also, Bouraiou et al. (2015) presented a modeling and simulation of single and double diode models using Matlab/Simulink. The accuracy of their proposed model was validated by experimental results for various values of irradiance and temperature. A novel single-diode model was also presented by Breitenstein (2014) for illuminated solar cells.

Ciulla et al. (2014) presented five PV panel models i.e., single, double and three diode models to describe the mathematical modeling of a PV cell. Moreover, these models were studied and analyzed in detail using an algorithm, which shows how the PV model parameters were extracted. Results from their study found out that parameters which define such components are directly related to the electrical behavior of the specific PV panel according to the weather conditions. Franzitta et al.

* Corresponding author.

E-mail address: engsalamjabr@gmail.com (S.J. Yaqoob).

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Nomenclatures

I	current of PV panel [A]
I_D	Shockley diode current [A]
I_{MPP}	Current of PV panel at MPP point [A]
I_O	Reverse saturation current of diode [A]
$I_{R_{SH}}$	Shunt-resistor current [A]
I_S	Saturation current of diode [A]
I_{SC}	Short-circuit current [A]
I_{PH}	photo-generated current [A]
$I_{PH,STC}$	photo-generated current at STC [A]
$I_{iss,i}$	Current of the issued models [A]
$I_{pro,i}$	Current of the proposed model [A]
G	Solar irradiance [W/m^2]
G_{STC}	Solar irradiance at STC [W/m^2]
K	constant of the Boltzmann [J/K]
K_I	temperature-current coefficient [$A/^{\circ}C$]
K_V	Voltage-temperature coefficient [$V/^{\circ}C$]
q	Electron charge [C]
R_S	Series resistance [Ω]
R_{SH}	Shunt resistance [Ω]
T_c	Cell Temperature [$^{\circ}C$]
T_{STC}	Cell Temperature at STC [$^{\circ}C$]
$MD(P)$	maximum difference in power [W]

$MD(I)$	maximum difference in current [A]
V	Voltage of PV panel [V]
V_{MPP}	Voltage of PV panel at MPP point [V]
V_{OC}	Open-circuit voltage [V]
$V_{pro,i}$	voltage the proposed model [V]
$V_{iss,i}$	voltage of the issued models [V]
N_S	Number of PV cells
P_{MPP}	Power of PV panel at MPP point [W]

Greek letter

α	Ideality factor of diode
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Abbreviations

AVCCS	Arbitrary voltage controlled current source
AVCVS	Arbitrary voltage controlled voltage source
DC	Direct current
DSP	Digital signal processing
FPGA	Field-programmable gate array
MPP	Maximum power point
PCB	printed circuit board
PIC	Peripheral interface
PV	Photovoltaic
PSIM	power simulation software
STC	Standard test conditions

(2016); Hovinen (1994) also used hypotheses, mathematical equations, and operative steps to extract single-diode circuit parameters with analytical techniques. Similarly, a new fast and accurate method for the extraction of single-diode circuit parameters based on datasheet parameters and experimental I–V curves was proposed by Laudani et al. (2014a). Their study reduced the complexity associated with mathematical considerations for extracting five parameters of PV cell or panel.

There exist extensive literature on this topic. Due to its simplicity, the single-diode model is one of the most used in the representation of the PV panel or cell equivalent circuit. This model is widely reported and extensively studied by Muhammadsharif et al. (2019). The main benefits of this model is that, it offers a good balance between accuracy and efficiency (Anani and Ibrahim, 2020). Furthermore, to characterize the PV panel, its I–V and P–V curves should be obtained and studied correctly under different environmental conditions with a suitable software (de Blas et al., 2002; De Soto et al., 2006). Therefore, Chatterjee et al. (2011) employed the Matlab m-file to show the I–V and P–V curves of a single-diode PV cell under different values of irradiance and temperature. Also, Ma et al. (2014a) studied and presented a simulation model for modeling a PV power generation system. The simulation results were obtained for different environmental conditions and compared with outdoor tests to show the I–V characteristics of the model. Additionally, a novel theoretical model was proposed by Ma et al. (2014b) to develop a simple and an accurate PV model. A PV module/string/array was developed in Matlab software to determine the PV model parameters and confirming them using experimental I–V curves at various values of irradiance and temperature.

Several researchers have used the Matlab/Simulink tool to model the mathematical equations of a PV circuit in order to study its characteristics and obtain the I–V and P–V graphs for different atmospheric conditions. A Matlab/Simulink based PV panel model includes an S-Function builder which is used to show the I–V and P–V characteristic of the PV panel under different environmental conditions (Ding et al., 2012). Yaqoob et al. (2021) proposed a new modeling method to demonstrate single and double diode models under various values of solar irradiance and temperature. This method was implemented using Matlab/Simulink based “Multiplexer and Functions blocks” that is included in the Matlab library. Besides, both single and double diode

models were simulated using Matlab/Simulink environment to show the output power and current characteristics of the PV cell under different intensity of solar irradiance and varying temperature (Hussain Basha et al., 2020). The simulation results indicated that the double diode model has higher efficiency compared to the single diode model. In addition, a step-by-step algorithm for modeling and simulating PV/cell/panel/array was proposed by (Nguyen and Nguyen, 2015). Moreover, a new method based on Tag tools in Matlab/Simulink software was used to build mathematical equations of a single-diode model under a wide range of weather conditions and physical parameters.

Other authors tried to obtain PV cell characteristics using a PSIM simulation tool. Chao et al. (2008) presented a PV panel model using PSIM software package. The proposed model was used to establish a PV system with rated power of 3 kW to obtain the I–V and P–V characteristics under normal and abnormal weather conditions. In addition, a mathematical PV model in PSIM software was modeled and simulated to obtain PV panel characteristics by El Hammoumi et al. (2018). A real time instrumentation using low-cost Arduino data acquisition for the PV model was used to validate the simulation results. On the other hand, a physical model simulator embedded in PSIM software was used to model and simulate a PV panel (Salman et al., 2015). A simple and accurate PV panel model was obtained, and its corresponding I–V and P–V curves were extracted for different environmental conditions. After which the simulation results were compared with that obtained from laboratory test results.

Actually, these tools do not contain in their libraries the electronic boards or micro-controllers (such as Arduino, FPGA, and DSP) in which PV system's applications (such as sun-tracker system, maximum power point tracking (MPPT) algorithm etc.) can be tested and practically implemented as it is in actual practical work (Singhal et al., 2011). As a result of the inequalities between software creation and real-world system requirements, many issues may occur in the realistic operation of the given system (Villalva et al., 2009). Moreover, the responsiveness of the real system's components may be different from those used in the simulation tool. So, this process may have an effect in the event of a bug and therefore, lead to an increase in the time spent in debugging runtime errors (Chalh et al., 2020). However, these issues can be solved by an electronic circuit design software called Proteus software tool.

Proteus has a printed circuit board (PCB), diagrammatic capture, and PROSPICE simulation layout models. The researcher can design and implement an algorithm by using many embedded boards that are included in the Proteus library such as Arduino, FPGA, and PIC. The controller algorithm can be written and uploaded directly on a micro-controller by a simple click on uploading hex code as it is similar to reality. Also, this tool is most popular because of the availability of electronic models or components that are used in practice, which can provide close results to practical implementation. Contrary to Proteus, in “Matlab/Simulink” or “PSIM” the user is required to rewrite an algorithm once the user starts the practical implementation. As a result, the researcher can use Proteus to check, analyze, and evaluate the system’s components or algorithms before the practical implementation of the work.

However, the Proteus software tool does not include a PV panel or cell model in its library. Yan et al. (2011) proposed a PV panel model in Proteus software. This model was created to show the I–V and P–V curves of the PV panel with a study on the partial shading effect on PV characteristics. The model was modeled using a voltage controller current source, a diode, a shunt resistor, and a series resistor. The drawback of this model is that it does not take into consideration the effect of weather conditions on the work of solar PV panel. Therefore, Chalh et al. (2020) presented a new PV panel model in Proteus using a single diode equivalent circuit model. This model was built in Proteus tool using a controlled current source and a diode with modified Spice code as a method to simulate the PV panel. However, the values of the diode parameters such as saturation diode current, number of PV panel cells, ideality constant, and energy of the band-gap were calculated at constant temperature $T = 25^\circ\text{C}$ and then written in the Spice code (motahhir et al., 2017). This model succeeded only in achieving the PV panel characteristics at different solar irradiance levels. Therefore, the user can change the irradiance value during the simulation by calculating the required photo generated current under standard test (STC) conditions of irradiance $G = 1000\text{W}/\text{m}^2$ and temperature $T = 25^\circ\text{C}$. Unfortunately, this model does not take into account the effect of changing the temperature on the characteristics of the PV panel (Saleh et al., 2020). Although this model was developed by Yaqoob and Obed (2019) using two-diode equivalent circuit PV panel, but the same problem remained without the possibility of varying the temperature during the simulation. Therefore, these models failed under different temperature conditions. This is a well-documented problem with these models (Chellakhi et al., 2021).

For this reason, the main novelty of this work is the methodology followed to model the PV panel based on its mathematical equations using a simple approach under Proteus for studying the effect of both irradiance and temperature, such a study has not been previously conducted to the best of authors’ knowledge. The PV panel is modeled based on the mathematical equations of a photocurrent source, a diode current and a shunt-resistor current for a single-diode equivalent circuit. Hence, a PV panel model can be tested and simulated under any environmental conditions, and thus a PV panel’s characteristics can easily be obtained. To validate this model results, the model is applied to two different types of polycrystalline silicon, and monocrystalline silicon PV panels. Additionally, the corresponding I–V and P–V curves were obtained for different values of irradiance and temperature. The plotted curves were then compared with the manufacturers data, PVsyst software Mermoud et al. (2012), and results of Villalva et al. (2009) and Chalh et al. (2020). Although it is true that the PV panel has been modeled mathematically under Simulink and PSIM, it should be mentioned that Proteus present a lot of advantages over MATLAB/Simulink and PSIM. Also, the proposed model can be used to test the performance of PV system application that contain the Arduino or PIC microcontrollers such as a MPPT algorithm, sun-tracker system, and solar inverter.

The research is presented as follows: section 2 covers the modeling of the PV panel circuit; the PV panel design in Proteus software is presented

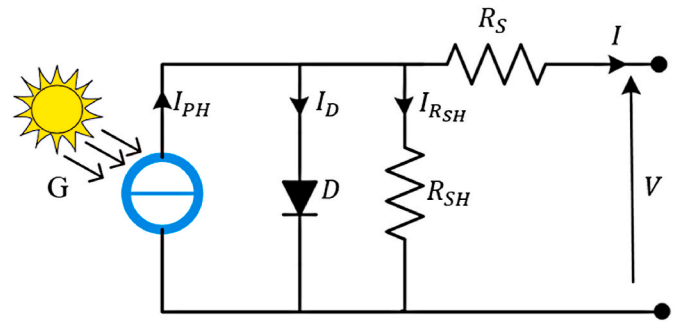


Fig. 1. Single -diode PV equivalent circuit.

in section 3. The results and discussion are presented in section 4, while the conclusion is presented in section 5.

2. Modeling of photovoltaic panel circuit

A PV panel device is essentially a semiconductor diode with a light-exposed P–N junction (Keevers and Green, 1994). A single diode equivalent circuit is adopted in this study, which is a simple, sufficient, and accurate way to represent the physical behavior of the PV device as seen in Fig. 1. This model includes a photo-generated current, a diode current, a series resistor, and a shunt resistor (de Blas et al., 2002; Ma et al., 2014b). Moreover, the total output current of the PV panel I is obtained using Kirchhoff's current law as shown in Eq. (1):

$$I = I_{PH} - I_D - I_{RSH} \quad (1)$$

where I_{PH} is the photo-generated current, I_D is the Shockley diode current, and I_{RSH} is the shunt-resistor current.

2.1. Photo-generated current

The photo-generated current, which represents the core of the PV panel in generating the required electricity is an essential part of the PV panel's model. It is directly proportional to solar irradiance and is also affected by the cell temperature according to Eq. (2) (Chalh et al., 2020; Villalva et al., 2009). As a result, this equation is used to model the photo-generated current in the Proteus software, it depends on these variables: irradiance G and temperature T as shown in Eq. (2):

$$I_{PH} = (I_{SC} - K_I (T - T_{STC})) \frac{G}{G_{STC}} \quad (2)$$

With respect to Eq. (2), it is important to note that the photo-generated current at STC conditions ($T_{STC} = 298.15\text{K}$, $G_{STC} = 1000\text{W}/\text{m}^2$) is approximately equal to the short-circuit current ($I_{PH,STC} \cong I_{SC}$). Also, this equation takes into account the cell temperature effect on the photo-generated current of ($K_I (T - T_{STC})$) where K_I is the temperature-current coefficient.

2.2. Shockley diode current

A PV model consist of many connected PV cells in series. Hence, the number of cells connected in series N_s should be incorporated into the Schlocky diode current. To describe the Schlocky diode equation, Eq. (3) should be defined for I_0 which is called the saturation diode current (Chellakhi et al., 2021; Villalva et al., 2009). Therefore, the Schlocky diode current of the PV panel can be described by Eq. (4) (Ding et al., 2012; Walker, 2001).

$$I_0 = \frac{I_{SC} + K_I (T - T_{STC})}{\exp\left(\frac{q(V_{OC} + K_V(T - T_{STC}))}{aK N_s T_c}\right) - 1} \quad (3)$$

Fig. 2. Model parameters of the SM55 PV panel using PV array tool.

$$I_D = I_o \left(\exp \frac{q(V + I R_S)}{a K N_s T} \right) \quad (4)$$

The terms of the above equations are defined as:

- α is the diode ideality factor
- K_V is the voltage temperature coefficient
- K is the constant of the Boltzmann ($1.380653 \times 10^{-23} \text{ J/K}$)
- q is the charge of the electron ($1.60217646 \times 10^{-19} \text{ C}$)
- N_s is the number of series cells
- R_S is the series resistance
- V is the output voltage of the PV panel V_{OC} is the open-circuit voltage of the PV panel

2.3. Shunt-resistor current

The resistance of the material and the cable links cause certain losses in the actual performance of the PV panel (Laudani et al., 2014a, 2014b). Hence, the series resistor R_S and shunt-resistor R_{SH} are applied to the mathematical PV model to represent these losses or a voltage drop (Lineykin et al., 2014). The R_S resistor does not represent the problem of the leaked current in the PV circuit model, which is in the range of milliohms. As a result, the R_{SH} resistor is considered to introduce the effect of this problem, and in general, it is high (Laudani et al., 2014b). The shunt-resistor current is considered in the total PV current equation to represent the leaked current as follows: Eqs. (5) and (6) (Ciulla et al., 2014).

$$I_{R_{SH}} = \frac{(V + I R_S)}{R_{SH}} \quad (5)$$

Therefore, the final total current of the PV panel can be rewritten as,

$$I = I_{PH} - I_S \left(\exp \left[\frac{(V + I R_S)}{A T} \right] - 1 \right) - \frac{(V + I R_S)}{R_{SH}} \quad (6)$$

$$\text{where } A = \frac{\alpha K N_s}{q}.$$

3. PV panel design in Proteus software

3.1. Extraction of the PV parameters

In order to build the proposed PV panel, the unknown parameters of the PV panel should be extracted. Therefore, in 2016, Mathworks released a new version of Matlab/Simulink version 2016, that include a

Table 1

Datasheet parameters of the SM55 and KC200GT panels at STC conditions.

Parameters	KC200GT	SM55
P_{MPP}	200W	55W
V_{MPP}	26.3V	17.4V
I_{MPP}	7.61A	3.15A
V_{OC}	32.9V	21.7V
I_{SC}	8.21A	3.45A
K_V	$-0.1230 \text{ V/}^\circ\text{C}$	$-0.077 \text{ V/}^\circ\text{C}$
K_I	$0.0032 \text{ A/}^\circ\text{C}$	$0.0012 \text{ A/}^\circ\text{C}$
N_s	54	36

Table 2

Extracted PV parameters using Matlab/PV array tool.

Parameters	KC200GT	SM55
R_S	0.335Ω	0.528Ω
R_{SH}	159.72Ω	134.64Ω
$I_{O,STC}$	$4.1746 \times 10^{-10} \text{ A}$	$9.0072 \times 10^{-11} \text{ A}$
α	1.0015	0.9645

PV array tool that allows one to quickly compute the internal unknown PV parameters by simply clicking on the PV parameters window. Once the manufacturer specifications are entered into this feature, it defines and shows the missing parameters as shown in Fig. 2. As discussed supra, the proposed model is applied on the SM55 monocrystalline, and the KC200GT multi-crystalline PV panels (Siemens Shell Monocrystalline Solar Array Datasheet. [Online]. Available: <http://www.atlantasolar.com/pdf/Shell/ShellSM55_USv1.pdf, and Kyocera Multi-crystal Photovoltaic Modules [Online]. Available: <<https://www.energymatters.com.au/images/kyocera/KC200GT.pdf>). Hence, their datasheet parameters at STC conditions are listed in Table 1. Also, Table 2 presents the extract parameters of these panels.

3.2. The influence of extracted parameters on the PV characteristics

The unknown PV parameters R_S , R_{SH} , $I_{O,STC}$, and α for both PV panels are estimated at STC conditions using the simple Matlab/PV array tool. However, the values of R_S , R_{SH} are affected on the I-V panel characteristics, as observed in Fig. 3, decreasing the R_{SH} changes the slope of

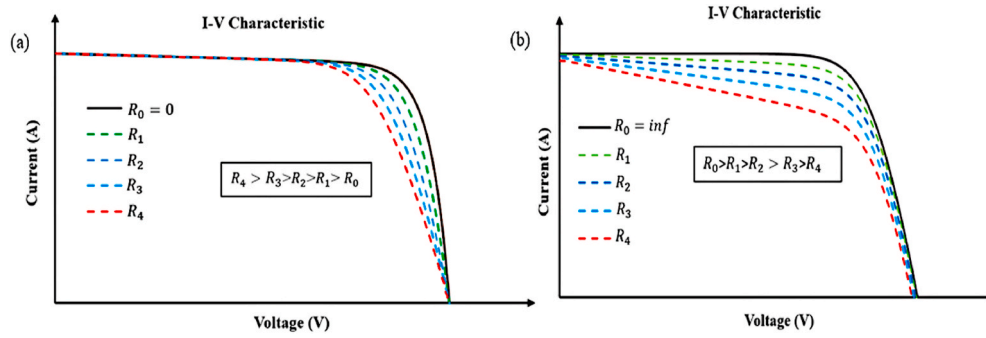


Fig. 3. (a) Series (b) shunt resistance effect on the I-V curve.

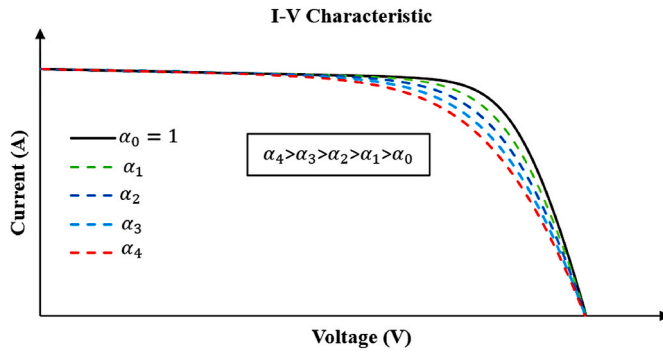


Fig. 4. Effect of diode ideality factor on the I-V characteristic.

the I-V characteristics in the upper part of current source region which moves till the MPP operating point. Also, an increase in R_s affect the shape of the I-V graph, especially in the lower section of the voltage source area and beyond the MPP. On the other hand, the effect of

ideality factor of the diode α appears on the “knee” area of the I-V characteristic graph as can be observed in Fig. 4. Unfortunately, the influence of these parameters can be seen at low irradiance and high-temperature values. Even if these parameters are not accurate, their impact on the PV curves is minor particularly for high irradiance levels. Because the main objective of the proposed paper is to present a new PV model in Proteus software, the best extraction method is not discussed here. For this reason, future researchers can use other precision methods to compute the unknown PV parameters to improve the accuracy of the plotted I-V and P-V curves.

3.3. Execution of the PV panel model in Proteus software

The theoretical equations of the photo-generated current, the diode current, and the shunt-resistor current are modeled using Proteus software. In fact, two steps were used to model the PV model, the first one involves the extraction of the internal PV panel parameters, and then integrating them in Proteus software, the second step is based on the modeled Eqs. (2)–(6) by using the component mode in Proteus software library as seen in Fig. 5.

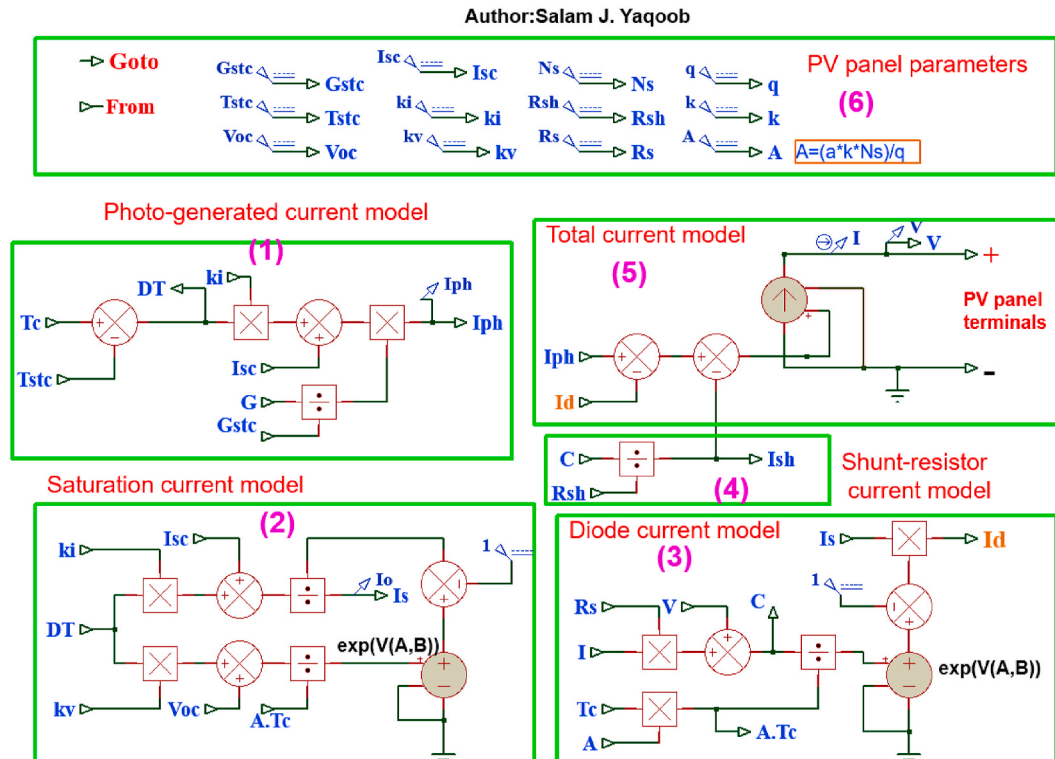


Fig. 5. Schematic diagram of the proposed PV model design in Proteus software.

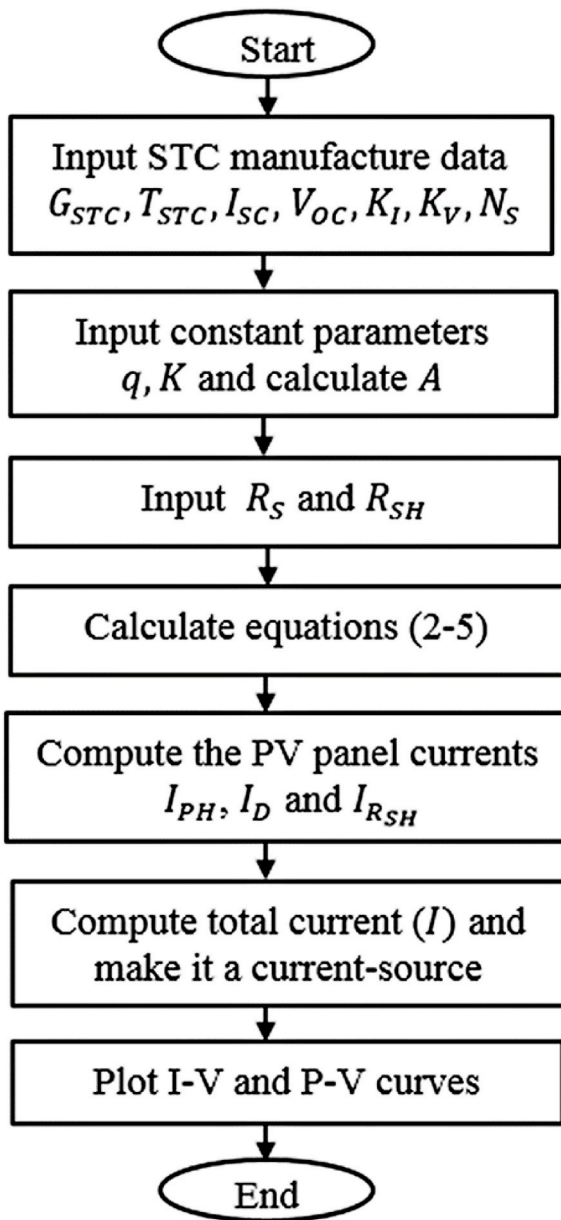


Fig. 6. Proposed model diagram.

Hence, this model provides an easy way to any user of the Proteus tool to build a PV panel model. Moreover, the proposed flowchart of this work is used to clarify the method as shown in Fig. 6. To simplify the proposed work, the model in Fig. 5 can be divided into six blocks to make it easier to comprehend as follows:

- Block (1): describes the photo-generated current model.
- Block (2): describes the saturation diode current model.
- Block (3): describes the diode current model.
- Block (4): describes the shunt-resistor current model.
- Block (5): describes the computation of the total PV current which is used as input to the arbitrary voltage controlled current source (AVCCS) device.
- Block (6): describes the datasheet and estimation parameters of the PV panel.

Besides, it is necessary to conduct the following steps in the Proteus tool to build a PV model:

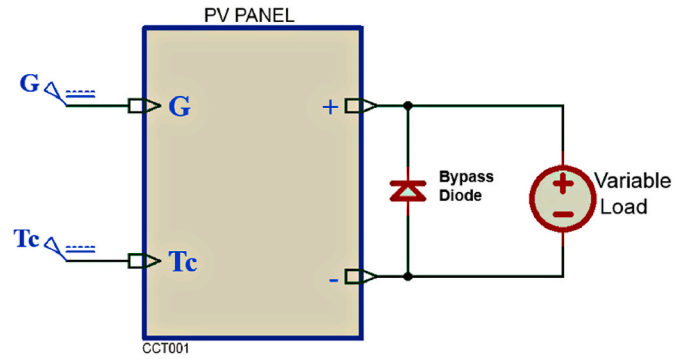


Fig. 7. The PV model sub-circuit.

- ✓ Launch the Proteus software application.
- ✓ Open the Schematic Capture.
- ✓ Click on Component Mode
- ✓ Open the Pick Devices.
- ✓ Open "Laplace Primitives".
- ✓ Select Laplace Addition, Subtraction, Multiplication, Division Operators, and then add them to your Schematic Capture window.
- ✓ Add the Arbitrary Voltage Controlled Voltage Source (AVCVS) and AVCCS to the Schematic Capture window by selecting "Modeling Primitives".
- ✓ Put the datasheet and estimation parameters values of the PV panel, and select "DC Generator" to set those values. After that, build the mathematical equations of the PV currents and connect your PV terminals to a bypass diode and a variable resistive load.
- ✓ Use "DC Voltage Source" as a variable load with value "X" for open-circuit voltage range of $(0 < V > V_{OC})$ to show the PV panel voltage axis in I-V and P-V curves.
- ✓ Use "DC SWEEP ANALYSIS" graph to see the simulation results and display the I-V and P-V curves of your PV panel.

In order to understand the modeling process, the tags for every parameter value is used which is presented in Proteus library. Next, a sub-circuit in the Proteus is used to simplify and form the PV panel model. As a result, a PV model with two inputs of irradiance and panel temperature and two output PV terminals of positive and negative signs as seen in Fig. 7 is obtained. However, the PV panel characteristics are affected by the change of irradiance and temperature. For this reason, the physical effect of these values on the P-V and I-V curves are investigated in this study. As observed, the photo-generated current which is modeled using Eq. (2) has the most effect on the output current of the PV panel. Therefore, the amount of this current is affected heavily by varying irradiance levels, which then affect the PV characteristics. To focus on this effect, the P-V and I-V curves are simulated under different irradiance conditions as will be shown in the next simulation results. Also, depending on Eq. (3), the diode saturation current is affected by changing the degree of the temperature. As a result, the output current of the PV panel increases exponentially as the level of temperature rises, while the voltage of the PV panel decreases linearly. The effect of varying the temperature is utilized also, in section 4. Therefore, this model allows the user to test the PV panel's characteristics under any weather condition i.e., irradiance and temperature.

4. Results and discussion

To verify the accuracy of the proposed model, both the I-V and P-V characteristics of the SM55 monocrystalline, and the KC200GT polycrystalline are obtained for different values of irradiance and temperature. The datasheet and extracted parameters of these panels under STC conditions are reported in the Proteus tool to build the model to obtain its plotted curves. Additionally, more exhaustive tests have been

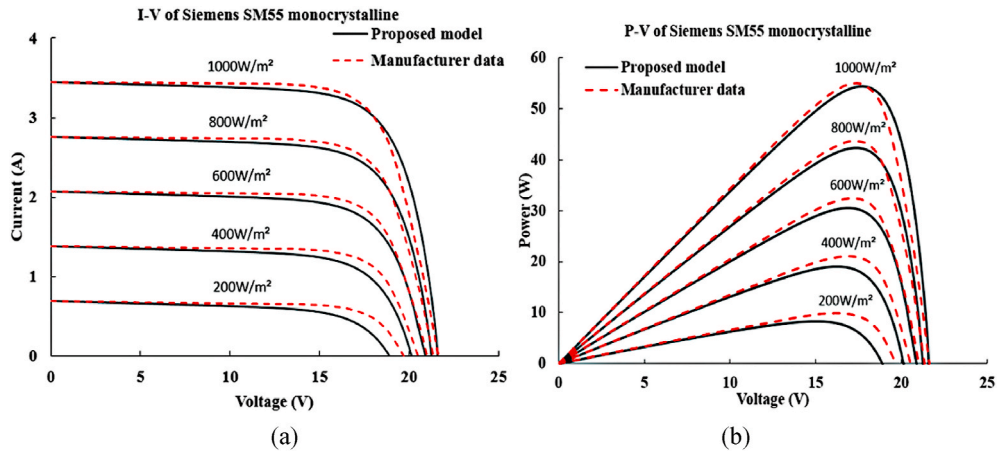


Fig. 8. (a) I-V (b) P-V curve comparison between the proposed model and manufacturer data at different irradiance values and constant temperature, $T = 25^{\circ}\text{C}$ for SM55 panel.

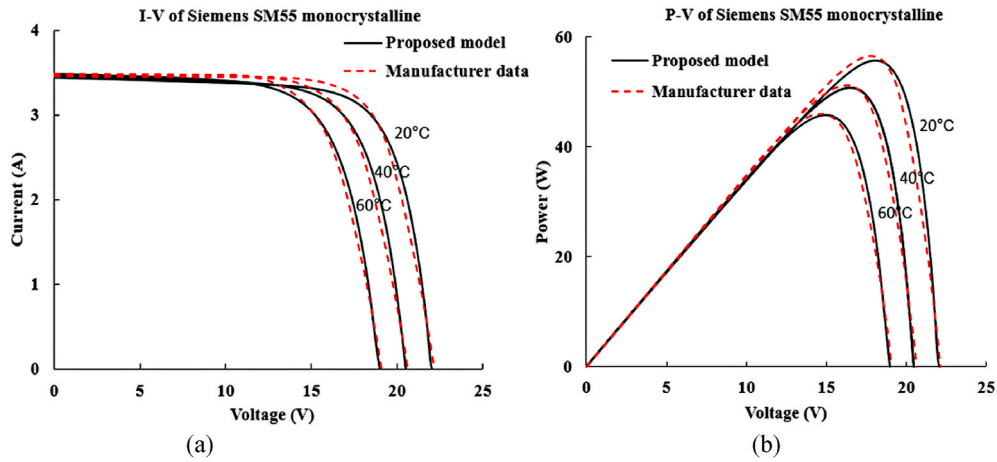


Fig. 9. (a) I-V (b) P-V curve comparison between the proposed model and manufacturer data at various temperature values and constant irradiance, $G = 1000\text{W}/\text{m}^2$ for SM55 panel.

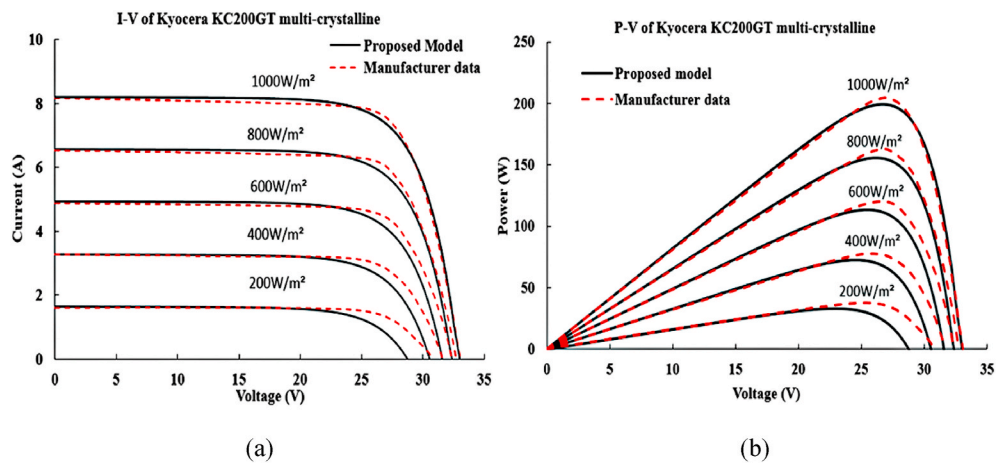


Fig. 10. (a) I-V (b) P-V curve comparison between the proposed model and manufacturer data at various levels of irradiance and constant temperature, $T = 25^{\circ}\text{C}$ for KC200GT panel.

conducted on the PV modules and then compared with those obtained by the manufacturer data, PVsyst software, and literature. Furthermore, the discussion is divided into three sections to study the performance and accuracy of the proposed model. In the first section, the plotted

curves of the SM55, and the KC200GT panels are compared with their manufacturer data. In the second section, we compared the proposed model's characteristics to those produced by the PVsyst software. On the other hand, a comparison of the plotted curves obtained from the

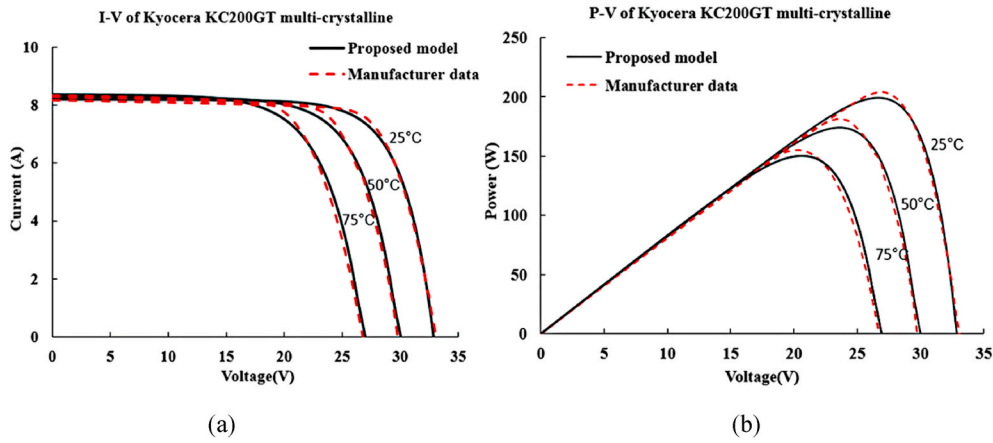


Fig. 11. (a) I–V (b) P–V curve comparison between the proposed model and manufacturer data at varying values of temperature and constant irradiance, $G = 1000 \text{ W/m}^2$ for KC200GT panel.

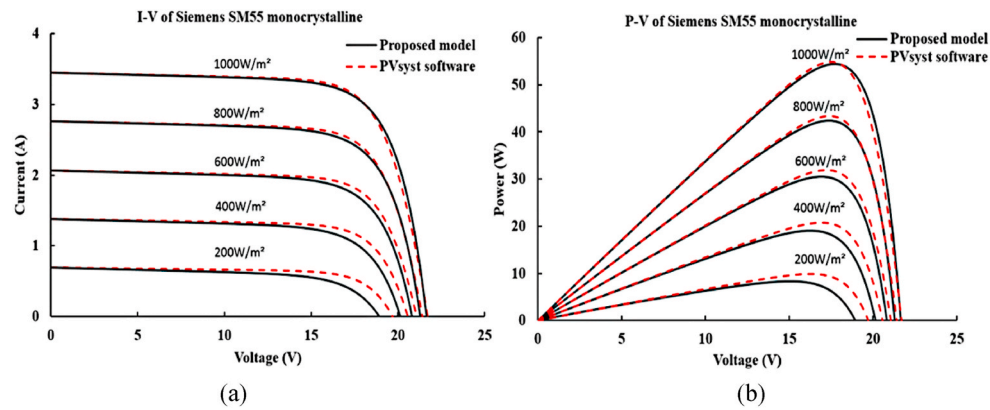


Fig. 12. (a) I–V (b) P–V graph comparison between the proposed model and PVsyst software for different levels of irradiance and constant temperature, $T = 25^\circ\text{C}$ for the SM55 panel.

proposed model and literature results curves are discussed in the third section.

4.1. Comparison with the manufacturer data

The plotted I–V and P–V curves of the proposed model are compared to curves of the manufacturer data under different atmospheric conditions. So, the plotted curves of the SM55 monocrystalline panel for

various irradiance and constant temperature $T = 25^\circ\text{C}$ are reported in Fig. 8. As can be observed, these figures show good agreement between the proposed model and manufacturer data with very small difference at low irradiance. Also, as demonstrated in Fig. 9, the plotted graphs fits well with the one provided by the manufacturer data when temperature is varied with constant irradiance $G = 1000 \text{ W/m}^2$.

Fig. 10 presents a comparison of the I–V and P–V curves for the KC200GT multi-crystalline panel between the proposed model and

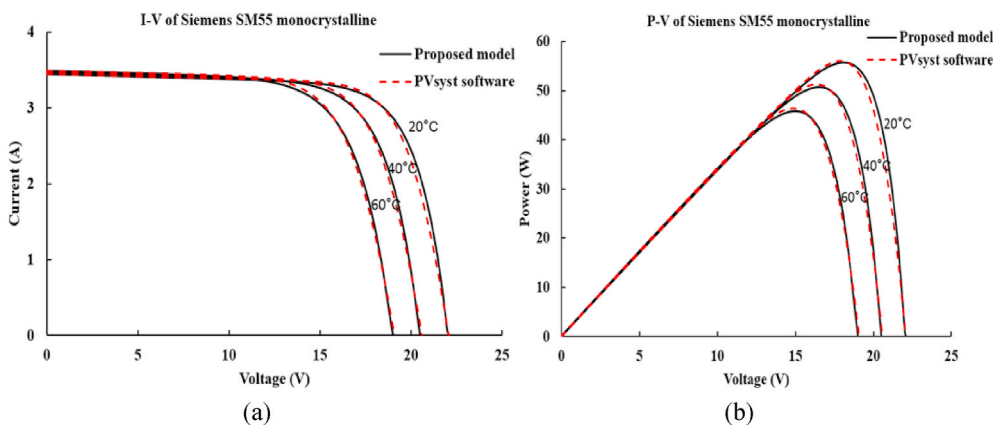


Fig. 13. (a) I–V (b) P–V graph comparison between the proposed model and PVsyst software for varying temperature and constant irradiance, $G = 1000 \text{ W/m}^2$ for the SM55 panel.

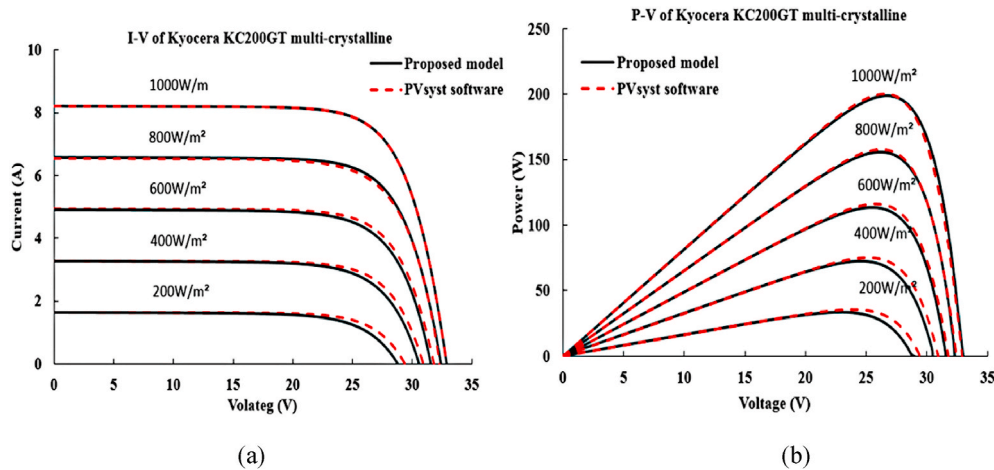


Fig. 14. (a) I-V (b) P-V graph comparison between the proposed model and PVsyst software curves for different levels of irradiance and constant temperature, $T = 25^{\circ}\text{C}$ for KC200GT panel.

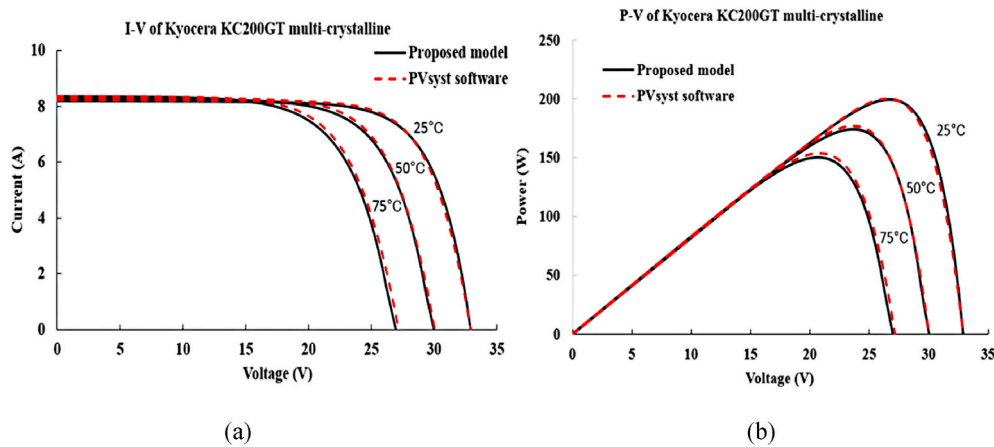


Fig. 15. (a) I-V (b) P-V graph comparison between the proposed model and PVsyst software for varying temperature and constant irradiance, $G = 1000\text{ W/m}^2$ for KC200GT panel.

manufacturer data for different values of irradiance and constant temperature, $T = 25^{\circ}\text{C}$. As seen, the plotted figures match with those given by manufacturer data for high irradiance levels, whereas only a slight disparity can be seen for low irradiance values. Besides, Fig. 11 illustrate the proposed model's curves for the KC200GT panel compared with the manufacturer's curves in case of different temperature values and constant irradiance $G = 1000\text{ W/m}^2$.

In summary, from these figures, the performance of the proposed model gave acceptable accuracy with negligible errors. Note, the flaw in the collected data from the manufacturer is the source of this disturbance in the curves. However, the proposed model's characteristics have good performance with a slight discordance around the MPP.

4.2. Comparison with the PVsyst software

The database of the PVsyst software is used to compare it with those of the proposed graphs to assess the performance of the proposed model. Fig. 12 presents the I-V and P-V graphs of the SM55 panel for various irradiance values and constant temperature $T = 25^{\circ}\text{C}$. As observed, the proposed model's curves for high irradiance levels agree well with those given by PVsyst software, whereas for low irradiance levels, only a small difference can be seen.

Fig. 13 show SM55 panel characteristics for different values of temperature and fixed irradiance $G = 1000\text{ W/m}^2$ compared with that of the PVsyst software. The proposed model's characteristics match those

of the PVsyst with little deviation around the MPP, but this is no influence.

Fig. 14 shows the comparison between the proposed model's curves and that of the PVsyst software for the KC200GT panel for different levels of irradiance and constant temperature $T = 25^{\circ}\text{C}$. As observed in these figures, the proposed model and PVsyst software results have identical graphs except for lower irradiance level of $G = 200\text{ W/m}^2$, at this irradiance there is a small discordance around the MPP. Also, the proposed model's curves for the KC200GT are compared with the one provided by PVsyst software for different values of temperature and fixed irradiance, as presented in Fig. 15. We obtained good performance of the proposed model characteristics with very minor deviations occurring at $T = 75^{\circ}\text{C}$.

Finally, because the database of those generated by the PVsyst was collected based on different unknown PV parameters such as ideality constant, series resistor, and shunt resistor they influenced the results. Nevertheless, the proposed model shows an acceptable accuracy with the PVsyst data under different environmental conditions.

4.3. Comparison with previously published results

In order to test the PV panel's characteristics, a comparison between the proposed model and already published works is done. The proposed I-V and P-V graphs for both PV panels are compared with models given by (Villalva et al., 2009), and (Chalh et al., 2020).

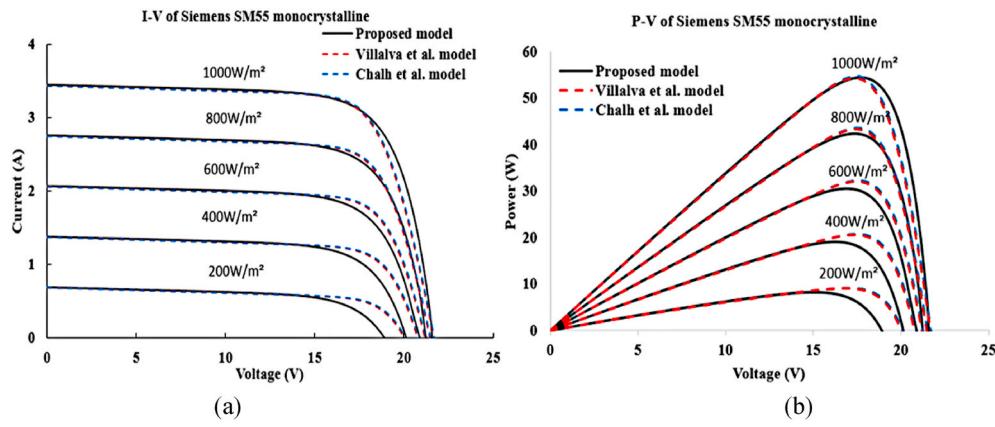


Fig. 16. (a) I-V (b) P-V graph comparison between the proposed model, Villalva et al. model, and (Chalh et al., 2020) model for different irradiance values and constant temperature, $T = 25^{\circ}\text{C}$ for the SM55 panel.

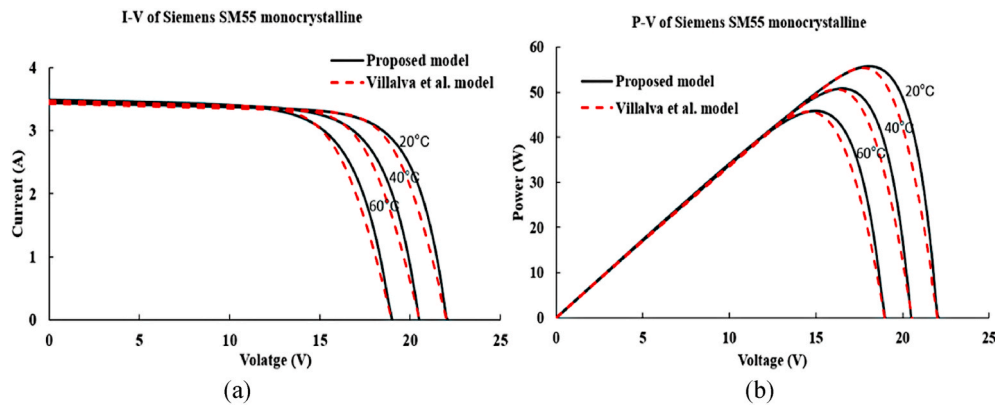


Fig. 17. (a) I-V (b) P-V graph comparison between the proposed model, and Villalva et al. (2009) model for varying temperature and fixed irradiance, $G = 1000\text{W}/\text{m}^2$ for SM55 panel.

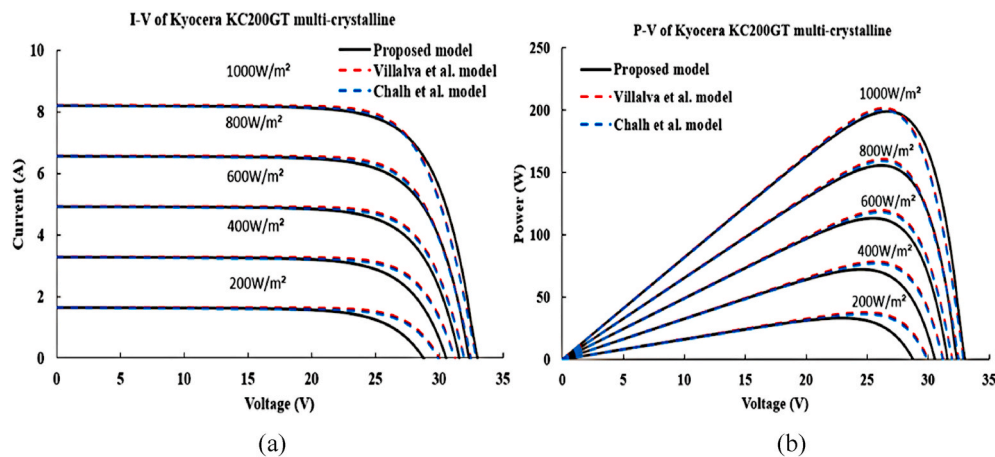


Fig. 18. (a) I-V (b) P-V graph comparison between the proposed model (Villalva et al., 2009), model, and (Chalh et al., 2020) model for different irradiance values and constant temperature, $T = 25^{\circ}\text{C}$ for the KC200GT panel.

Fig. 16 illustrate the comparison of SM55 panel curves for different levels of irradiance and constant temperature $T = 25^{\circ}\text{C}$. It can be seen that the proposed model performs well with the results obtained by (Villalva et al., 2009) and (Chalh et al., 2020) models. The results of (Villalva et al., 2009) and (Chalh et al., 2020) models appear to be relatively similar at first glance, but closer inspection reveals minor variances, most likely because they used a more accurate unknown

parameters in the simulation.

As shown for high irradiance levels of more than $600\text{W}/\text{m}^2$, the proposed model shows an acceptable performance. In addition, a small difference can be seen for irradiance less than $600\text{W}/\text{m}^2$. Also, the I-V and P-V graphs for temperatures ($T = 20^{\circ}\text{C}$, 40°C , and 60°C) and for $G = 1000\text{W}/\text{m}^2$ are reported, respectively in Fig. 17. In this figures, we compared this study's data with that of Villalva et al. (2009) model only,

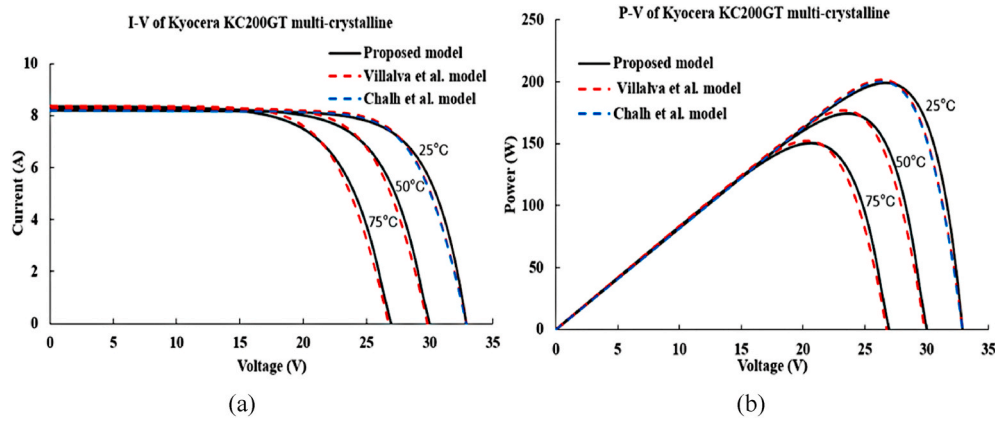


Fig. 19. (a) I–V (b) P–V graph comparison between the proposed model and Villalva et al. (2009) model, and (Chalh et al., 2020) model for varying temperature and constant irradiance, $G = 1000\text{W/m}^2$ for KC200GT panel.

Table 3

Differences in current at MPP between the proposed model and the issued I–V curves of SM55 panel at different irradiances and fixed temperature $T = 25^\circ\text{C}$.

PV parameters at different MPP points		Irradiance (W/m^2)				
		200	400	600	800	1000
Manufacturer data	Voltage (V)	16.20	17.16	17.15	17.10	17.27
	Current (A)	0.61	1.23	1.89	2.56	3.19
	Proposed	0.55	1.175	1.797	2.456	3.069
	Current (A)					
	Difference (A)	<u>0.06</u>	<u>0.055</u>	<u>0.093</u>	<u>0.104</u>	<u>0.121</u>
PVsyst software	Voltage (V)	16.222	16.877	17.20	17.389	17.469
	Current (A)	0.604	1.227	1.858	2.494	3.138
	Proposed	0.55	1.175	1.797	2.456	3.069
	Current (A)					
	Difference (A)	0.054	0.052	0.061	0.038	0.069
Villalva et al. model	Voltage (V)	16.85	17.35	17.45	17.45	17.35
	Current (A)	0.539	1.187	1.838	2.483	3.13
	Proposed	0.55	1.175	1.797	2.456	3.069
	Current (A)					
	Difference (A)	–0.011	0.012	0.041	0.027	0.061
Chalh et al. model	Voltage (V)	17	17.48	17.49	17.5	17.5
	Current (A)	0.538	1.185	1.84	2.49	3.12
	Proposed	0.55	1.175	1.797	2.456	3.069
	Current (A)					
	Difference (A)	–0.012	0.01	0.043	0.034	0.051

The underline values indicates the highest value of $MD(I)$ for each irradiance.

this is because the Chalh et al. (2020) model fails at any temperature value except $T = 25^\circ\text{C}$. It is for this reason that this model is proposed, as demonstrated in the proposed results the plotted curves achieved through the proposed model match with those obtained by the Villalva et al. (2009) model except for a very small deviations.

Fig. 18 gives the I–V and P–V comparison between the proposed model, Villalva et al. (2009) and Chalh et al. (2020) models for the KC200GT panel under various levels of irradiance and fixed temperature $T = 25^\circ\text{C}$. From these figures, the proposed model have good accuracy except for lower irradiance level of $G = 200\text{W/m}^2$, at this irradiance there is a small discordance around the MPP.

Fig. 19 gives I–V and P–V characteristics for comparison between the proposed work and those obtained by Villalva et al. (2009) and Chalh et al. (2020) models for temperature ranging from 25°C to 75°C and $G = 1000\text{W/m}^2$. As seen, the proposed model gives accurate results and agrees well with that obtained by Villalva et al. (2009) model.

Furthermore, in order to assess the accuracy and dispersion of the proposed model results, the maximum difference (MD) at MPP for the

Table 4

$MD(I)$ s at MPP between the proposed and the issued I–V graphs of the SM55 panel at different temperatures and fixed irradiance $G = 1000\text{W/m}^2$.

PV parameters at different MPP points		Temperature ($^\circ\text{C}$)		
		20	40	60
Manufacturer data	Voltage (V)	17.60	16.04	14.79
	Current (A)	3.21	3.19	3.11
	Proposed Current (A)	3.095	3.075	3.052
	Difference (A)	<u>0.115</u>	<u>0.115</u>	0.058
	Voltage (V)	17.857	16.34	14.845
PVsyst software	Current (A)	3.137	3.133	3.122
	Proposed Current (A)	3.095	3.075	3.052
	Difference (A)	0.042	0.058	0.07
Villalva et al. model	Voltage (V)	17.73	16.16	14.64
	Current (A)	3.128	3.134	3.126
	Proposed Current (A)	3.095	3.075	3.052
	Difference (A)	0.033	0.059	<u>0.074</u>
Chalh et al. model	Voltage (V)	×	×	×
	Current (A)	×	×	×
	Proposed Current (A)	3.095	3.075	3.052
	Difference (A)	×	×	×

The underline values indicates the highest value of $MD(I)$ for each temperature.

current and power is calculated using the following expressions:

$$MD(I) = \text{MAX}[I_{pro,i} - I_{iss,i}] \quad (7)$$

$$MD(P) = \text{MAX}[V_{pro,i} I_{pro,i} - V_{iss,i} I_{iss,i}] \quad (8)$$

where $I_{pro,i}$ and $V_{pro,i}$ are the current and voltage of the i -th point obtained from the proposed model, while $I_{iss,i}$ and $V_{iss,i}$ are the current and voltage of the i -th point obtained from the I–V curves issued by other studies.

Table 3 reports the maximum current differences $MD(I)$ s in I–V characteristics of the SM55 panel at different values of irradiance and fixed temperature $T = 25^\circ\text{C}$. As observed, the smallest current differences are observed for the Villalva et al. (2009) model which seems more accurate with respect to the proposed model; the $MD(I)$ s vary from -0.11 A to 0.061 A . On the other hand, the highest values of $MD(I)$ s are seen for the manufacturer data which are reported from 0.06 A at $G = 200\text{W/m}^2$ to 0.121 A at $G = 1000\text{W/m}^2$.

Table 4 presents the $MD(I)$ s results for I–V curves of the SM55 panel at different temperatures and constant irradiance $G = 1000\text{W/m}^2$. Considering I–V curves at different temperatures, the PVsyst software shows the smallest $MD(I)$ s values which are contained in the range from 0.042 A to 0.07 A . As seen in this Table, the proposed model succeeded in obtaining I–V curves at different temperatures with acceptable accuracy

Table 5Differences in power at MPP between the proposed and the issued P–V curves of the SM55 panel at different irradiances and fixed temperature. $T = 25^{\circ}\text{C}$

PV parameters at different MPP points		Irradiance (W/m^2)				
		200	400	600	800	1000
Manufacturer data	Voltage (V)	16.20	17.16	17.15	17.10	17.27
	Power (W)	9.879	21.059	32.445	43.712	55.006
	Proposed Power (W)	8.256	19.107	30.554	42.376	54.476
	Difference (W)	<u>1.623</u>	<u>1.952</u>	<u>1.891</u>	<u>1.336</u>	<u>0.53</u>
PVsyst software	Voltage (V)	16.222	16.877	17.20	17.389	17.469
	Power (W)	9.8	20.718	31.96	43.379	54.82
	Proposed Power (W)	8.256	19.107	30.554	42.376	54.476
	Difference (W)	1.544	1.611	1.406	1.003	0.344
Villalva et al. model	Voltage (V)	16.85	17.35	17.45	17.45	17.35
	Power (W)	9.084	20.607	32.078	43.335	54.314
	Proposed Power (W)	8.256	19.107	30.554	42.376	54.476
	Difference (W)	0.828	1.5	1.524	0.959	−0.162
Chalh et al. model	Voltage (V)	17	17.48	17.49	17.5	17.5
	Power (W)	9.152	20.75	32.284	43.601	54.62
	Proposed Power (W)	8.256	19.107	30.554	42.376	54.476
	Difference (W)	0.896	1.643	1.73	1.225	0.144

The underline values indicates the highest value of $MD(P)$ for each irradiance.**Table 6**Differences in power at MPP between the proposed and the issued P–V curves of SM55 panel at different values of temperature and fixed irradiance $G = 1000\text{W}/\text{m}^2$.

PV parameters at different MPP points		Temperature ($^{\circ}\text{C}$)		
		20	40	60
Manufacturer data	Voltage (V)	17.60	16.04	14.79
	Power (W)	56.51	51.127	45.947
	Proposed Power (W)	55.715	50.748	45.784
	Difference (W)	<u>0.795</u>	0.379	0.163
PVsyst software	Voltage (V)	17.857	16.34	14.845
	Power (W)	56.019	51.206	46.349
	Proposed Power (W)	55.715	50.748	45.784
	Difference (W)	0.304	<u>0.458</u>	<u>0.565</u>
Villalva et al. model	Voltage (V)	17.73	16.16	14.64
	Power (W)	55.468	50.646	45.769
	Proposed Power (W)	55.715	50.748	45.784
	Difference (W)	−0.247	−0.102	−0.015
Chalh et al. model	Voltage (V)	×	×	×
	Power (W)	×	×	×
	Proposed Power (W)	55.715	50.748	45.784
	Difference (W)	×	×	×

The underline values indicates the highest value of $MD(P)$ for each temperature.**Table 8**Differences in current at MPP between the proposed and the issued I–V curves of KC200GT panel at different values of temperature and fixed irradiance $G = 1000\text{W}/\text{m}^2$.

PV parameters at different MPP points		Temperature ($^{\circ}\text{C}$)		
		25	50	75
Manufacturer data	Voltage (V)	26.84	23.31	20.06
	Current (A)	7.62	7.77	7.74
	Proposed Current (A)	7.484	7.435	7.287
	Difference (A)	0.136	<u>0.335</u>	<u>0.453</u>
PVsyst software	Voltage (V)	26.484	23.626	20.778
	Current (A)	7.560	7.500	7.418
	Proposed Current (A)	7.484	7.435	7.287
	Difference (A)	0.076	0.065	0.130
Villalva et al. model	Voltage (V)	26.363	23.276	20.273
	Current (A)	7.643	7.597	7.514
	Proposed Current (A)	7.484	7.435	7.287
	Difference (A)	<u>0.159</u>	0.162	0.227
Chalh et al. model	Voltage (V)	26.25	×	×
	Current (A)	7.622	×	×
	Proposed Current (A)	7.484	7.435	7.287
	Difference (A)	0.138	×	×

The underline values indicates the highest value of $MD(I)$ for each temperature.**Table 7**Differences in current at MPP between the proposed and the issued I–V curves of KC200GT panel at different irradiances and fixed temperature. $T = 25^{\circ}\text{C}$

PV parameters at different MPP points		Irradiance (W/m^2)				
		200	400	600	800	1000
Manufacturer data	Voltage (V)	25.34	25.25	26.66	26.74	26.84
	Current (A)	1.48	3.08	4.51	6.1	7.62
	Proposed Current (A)	1.462	2.952	4.437	5.932	7.484
	Difference (A)	0.018	<u>0.128</u>	0.073	0.168	0.136
PVsyst software	Voltage (V)	23.827	25.0855	25.7548	26.1843	26.4844
	Current (A)	1.4956	3.0056	4.5211	6.0402	7.5605
	Proposed Current (A)	1.462	2.952	4.437	5.932	7.484
	Difference (A)	0.0336	0.0536	0.0841	0.1082	0.0765
Villalva et al. model	Voltage (V)	24.814	25.69	26.083	26.278	26.363
	Current (A)	1.522	3.056	4.589	6.118	7.6439
	Proposed Current (A)	1.462	2.952	4.437	5.932	7.484
	Difference (A)	<u>0.06</u>	0.104	<u>0.152</u>	<u>0.186</u>	<u>0.159</u>
Chalh et al. model	Voltage (V)	24.85	25.55	25.9	26.25	26.25
	Current (A)	1.468	3.019	4.566	6.071	7.622
	Proposed Current (A)	1.462	2.952	4.437	5.932	7.484
	Difference (A)	0.006	0.067	0.129	0.139	0.138

The underline values indicates the highest value of $MD(I)$ for each irradiance.

Table 9Differences in power at MPP between the proposed and the issued P–V curves of KC200GT panel at different values of irradiance and fixed temperature. $T = 25^{\circ}\text{C}$

PV parameters at different MPP points		Irradiance (W/m^2)				
		200	400	600	800	1000
Manufacturer data	Voltage (V)	25.34	25.25	26.66	26.74	26.84
	Power (W)	37.503	77.77	120.236	163.114	204.52
	Proposed Power (W)	33.264	72.307	113.365	155.731	199.077
	Difference (W)	4.239	5.463	<u>6.871</u>	<u>7.383</u>	<u>5.443</u>
PVsyst software	Voltage (V)	23.827	25.0855	25.7548	26.1843	26.4844
	Power (W)	35.635	75.396	116.44	158.158	200.235
	Proposed Power (W)	33.264	72.307	113.365	155.731	199.077
	Difference (W)	2.371	3.089	3.075	2.427	1.158
Villalva et al. model	Voltage (V)	24.814	25.69	26.083	26.278	26.363
	Power (W)	37.778	78.532	119.703	160.778	201.518
	Proposed Power (W)	33.264	72.307	113.365	155.731	199.077
	Difference (W)	<u>4.514</u>	<u>6.225</u>	6.338	5.047	2.441
Chalh et al. model	Voltage (V)	24.85	25.55	25.9	26.25	26.25
	Power (W)	36.494	77.156	118.265	159.37	200.095
	Proposed Power (W)	33.264	72.307	113.365	155.731	199.077
	Difference (W)	3.23	4.849	4.9	3.639	1.018

The underline values indicates the highest value of $MD(P)$ for each irradiance.

Table 10Differences in power at MPP between the proposed and the issued P–V curves of KC200GT panel at different temperatures and fixed irradiance $G = 1000\text{W}/\text{m}^2$.

PV parameters at different MPP points		Temperature ($^{\circ}\text{C}$)		
		25	50	75
Manufacturer data	Voltage (V)	26.84	23.31	20.06
	Power (W)	204.52	180.977	155.1967
	Proposed Power (W)	199.077	174.35	150.49
	Difference (W)	<u>5.443</u>	<u>6.627</u>	<u>4.706</u>
PVsyst software	Voltage (V)	26.484	23.626	20.778
	Power (W)	200.23	177.22	154.13
	Proposed Power (W)	199.077	174.35	150.49
	Difference (W)	1.153	2.87	3.64
Villalva et al. model	Voltage (V)	26.363	23.276	20.273
	Power (W)	201.51	176.83	152.341
	Proposed Power (W)	199.077	174.35	150.49
	Difference (W)	2.433	2.48	1.851
Chalh et al. model	Voltage (V)	26.25	×	×
	Power (W)	200.09	×	×
	Proposed Power (W)	199.077	174.35	150.49
	Difference (W)	1.013	×	×

The underline values indicates the highest value of $MD(P)$ for each temperature.

whiles the Chalh et al. (2020) model failed under this test. Moreover, from the maximum power differences of the SM55 panel as reported in Tables (5 and 6) it is clear that the proposed model shows a good accuracy. As observed, the smallest $MD(P)$ s at fixed temperature and varying irradiance are again obtained and compared to the Villalva et al. (2009) model that produces $MD(P)$ s varying from -0.162W at $G = 1000\text{W}/\text{m}^2$ to 1.524W at $G = 600\text{W}/\text{m}^2$. On the other hand, the greatest $MD(P)$ s in P–V characteristics at different temperatures are generated in the PVsyst software which are contained from 0.458W at $T = 40^{\circ}\text{C}$ to 0.562W at $T = 60^{\circ}\text{C}$. Thus, a little discordance in maximum power value exist between the proposed model and the PVsyst software. Nevertheless, this small difference in MPP is considered not so important because the proposed model has passed the test of a PV panel when it is exposed to a sudden change in temperature.

Table 7 shows the current differences at MPP in I–V graphs of the KC200GT panel under different irradiances and constant temperature $T = 25^{\circ}\text{C}$. As seen in this Table, the greatest $MD(I)$ s result are shown in the Villalva et al. (2009) model that yields $MD(I)$ s varying from 0.06A to 0.186A . With this difference in currents in the proposed and Villalva et al. (2009) models, the proposed model presents a good performance compared to other models. In addition, the $MD(I)$ s between the proposed and the issued I–V characteristics for same panel under different

temperatures are reported in Table 8. As seen, the Chalh et al. (2020) model did not pass through $T = 50^{\circ}\text{C}$ and 75°C , because of the fact that the Spice code of the diode was created at constant temperature $T = 25^{\circ}\text{C}$. Also, we recorded good agreement between the proposed model results, PVsyst software, manufacturer data, and the Villalva et al. (2009) model, except for high temperatures above 50°C , in the case of manufacturer data this disturbance is caused by the error of the collected data.

Table 9 presents the power differences at MPP in P–V graphs of the KC200GT panel for different irradiances and constant temperature $T = 25^{\circ}\text{C}$. As present in the Table, the maximum difference in power occurred in the case of comparison with the manufacturer data which has a yield $MD(P)$ of about 7.38W at $G = 800\text{W}/\text{m}^2$.

Table 10 gives the comparison between the proposed and issued P–V curves under different temperatures. Moreover, small differences between the proposed current and other models occurred particularly under higher irradiance conditions. It is important to note that the Chalh et al. (2020) model was tested under only $T = 25^{\circ}\text{C}$. As observed, an acceptable $MD(P)$ value of about 1W at $G = 100\text{W}/\text{m}^2$ is obtained, where the proposed model presents a good accuracy when the temperature increased from 25°C to 75°C .

5. Conclusion

In this paper a new PV panel model is developed and proposed using Proteus software tool. A single-diode circuit which consist of a photo-generated current, a shunt resistor, and a series resistor is used. First, the theoretical analysis of the single-diode circuit is analyzed and then the mathematical equations of the photo-generated current, the diode current, and the shunt-resistor current are obtained. And thereafter, these equations were modeled using the “Laplace Primitives Operators” and “Modeling Primitives” tools in “Pick Devices option” of the Proteus software library. Second, the unknown internal PV parameters are extracted using a PV array tool in Matlab/Simulink. And the effect of these parameters on the performance and accuracy of the proposed model is considered.

The performance is tested by comparing the proposed model results with manufacturer data, PVsyst tool, and to results in previously published works under arbitrary environmental conditions. These comparisons have proven that the proposed model has a good accuracy in obtaining I–V characteristics. However, the observed slight deviations between the proposed model and that of the manufacturer’s data are as a result of the flaws in the manufacturer’s data, this accounted for the differences in the curves. Therefore, it’s possible to deduce that, the proposed model is simple and useable to verify several types of solar PV

panels.

This study therefore recommend the following as future research areas: (1) study of the partial shading PV condition using Proteus software; and (2) Design and simulation of a MPPT algorithm using the proposed PV simulator.

CRedit authorship contribution statement

Salam J. Yaqoob: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Software, Data curation. **Saad Motahhir:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Software, Data curation. **Ephraim Bonah Agyekum:** Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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